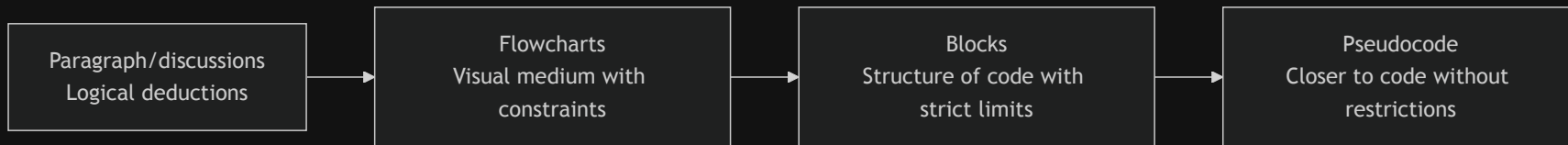


Blocks

Progression of complexity

Programming *can* be quite complex, in order to traverse that complexity and only care about the concepts one at a time



Blocks and Scratch

When talking about blocks, the primary tool being used is

scratch

A little bit of history:

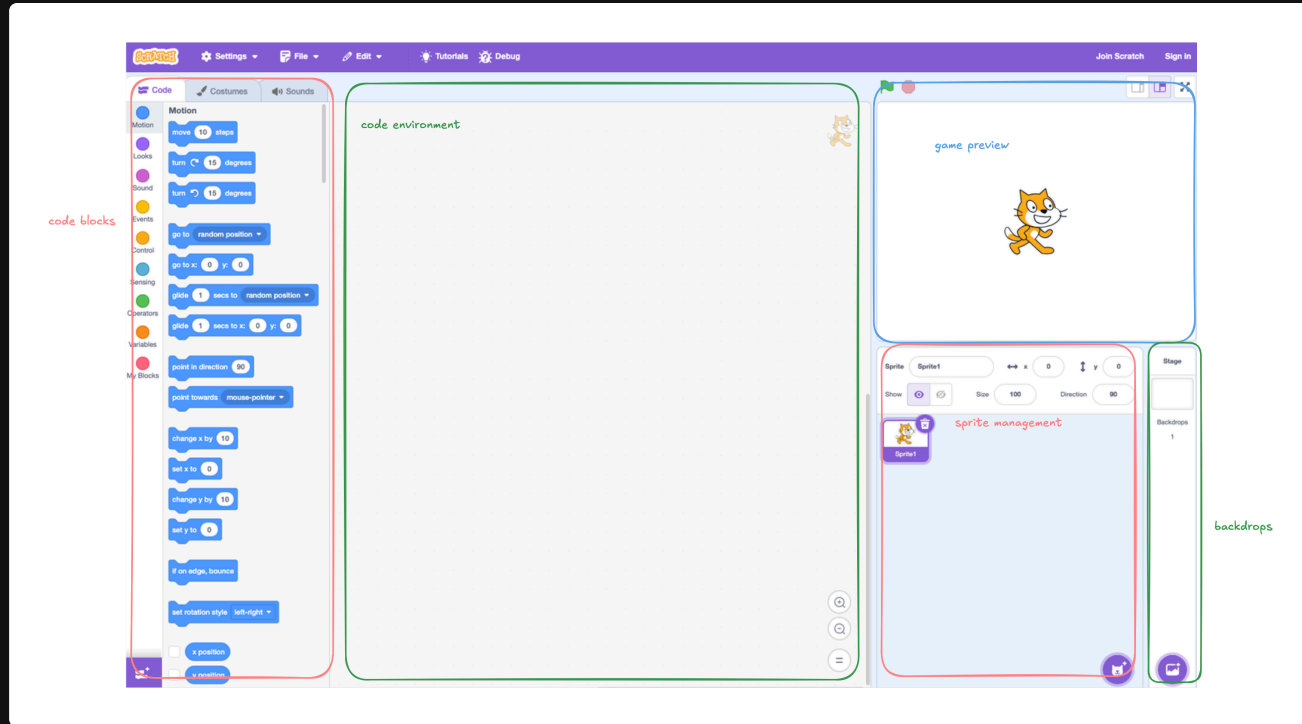
- scratch is a block based programming language and website created in 2003
- It was made by the MIT Media Lab for the purposes of education

Today it's a popular game engine, especially for beginner programmers

demo

other demo

Scratch Interface



Fundamental Concepts in Scratch

Code Blocks

There are 9 different block **categories** with different *colors*

of note is the *start block*

`when [flag] clicked` which is used at the top of a programming **stack**

To create a program, drag blocks from the *palette* to the *blank space*, and *interlock* blocks together

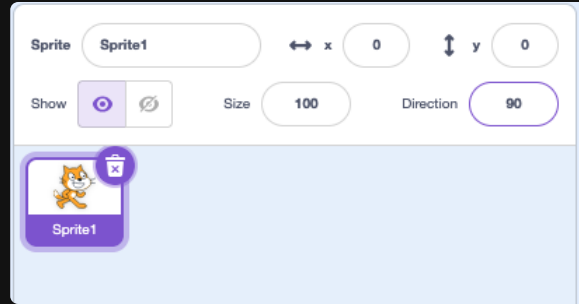
First program

https://scratch-tutorial.readthedocs.io/fr/latest/1_intro/intro.html#your-first-program

Sprites

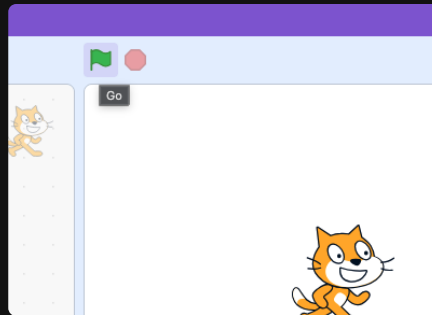
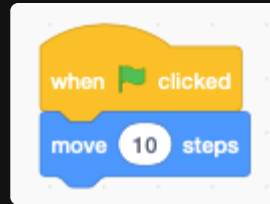
Sprites are *objects* in the game, these objects have *properties*

In particular, a *direction*, an *x* and a *y* position



Moving the sprite

1. Select the *events* category
2. Drag the "when [flag] clicked"
3. select the *motion* category
4. Drag the *move 10 steps* block to the canvas and attach it



Interacting with the sprite

1. In the control category, drag the *when space key pressed* block to the board
2. In the motion category, add a motion below the press

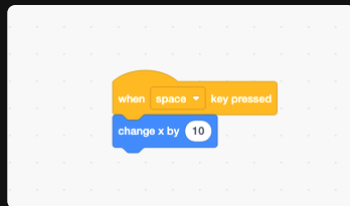


Challenge number 1

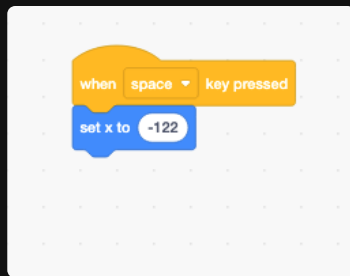
Make a game where if I press left and right, I move left or right

Other ways of moving

1. Add another if key pressed block
2. In the motion category, select the *change x by* block



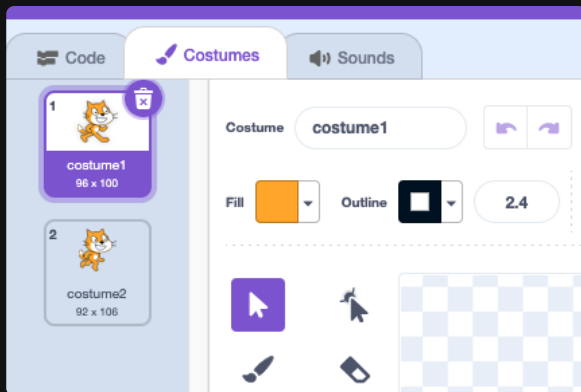
1. Add another if key pressed block
2. In the motion category, select the *set x to* block



Costumes in scratch

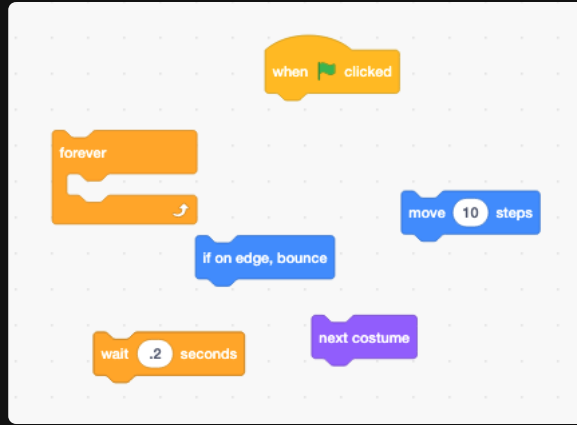
Every *sprite* can have multiple *costumes*

Otherwise known as *frames*



given that in the **looks** category, there's a *next costume* block, make your sprite "*walk*", whenever it moves

Challenge number 1



Given these blocks, make the cat

1. move when you start the game (with the flag)
2. bounce on the edge of the screen
3. have a walk animation
4. have a 0.2 second delay

Note, set the initial direction of the sprite to not be 90 degrees